

WINDMILL MANUFACTURING COMPANY AT THOOTHUKUDI

COMPREHENSIVE TECHNICAL REPORT

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2026

2. Scope of the Report

The report covers the proposed manufacturing facility from the initial land concept through production and dispatch. It includes the approximately 12-acre land assumption, building-area schedule, machine categories, blade manufacturing systems, assembly and testing, employee structure, transport routes and costs, capital investment, annual operating cost, production and revenue assumptions, quality controls, safety considerations and future digital improvements.

The supplied material also presents a web-based information layer at windweb.netlify.app. In this report the website is treated as the digital presentation associated with the manufacturing concept. The report does not invent undocumented software functions, databases, monitoring systems or operational claims. Website-specific statements are limited to what is supported by the supplied material.

The scope is therefore broad enough to describe the complete industrial system, but the document continues to distinguish conceptual planning from detailed engineering. Certified structural drawings, detailed aerodynamic blade design, statutory approval packages, confirmed customer contracts and verified vendor quotations are outside the evidence supplied for this report.

Engineering interpretation and control points

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3. Executive Summary

The proposed facility is located conceptually at Thoothukudi, Tamil Nadu, with an approximate land requirement of 12 acres. The presentation gives a land-cost range of ₹144 crore to ₹540 crore and uses ₹300 crore as the selected average in the financial model. Building cost is presented as ₹13–26 crore, while the selected financial model uses ₹20 crore. The machine-investment range is presented as ₹80–150+ crore for a large-scale factory, with ₹100 crore used in the selected model.

The workforce model in the presentation is 450–550 employees. Annual operating expenditure is shown as ₹175 crore, consisting of salary, maintenance, electricity and utilities, transport and raw materials. Production is assumed at 100 blades per year, with a selling-price range of ₹3–5 crore per blade and an average assumption of ₹4 crore. This gives illustrative annual revenue of ₹400 crore. The presentation then subtracts ₹175 crore annual expense and reports an arithmetic difference of ₹225 crore.

These values are useful for organizing the concept but should not be interpreted as audited profit, investment return or commercial performance. A full financial model would additionally address depreciation, financing, tax, working capital, ramp-up, rejection, downtime, insurance, administration and other costs.

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4. Report Basis and Evidence

The primary evidence for the numerical sections is the supplied 32-slide presentation, supported by the earlier report document. The presentation explicitly provides tables for building areas, machinery, employees, blade categories, transportation, investment, operating expenditure, production and the final arithmetic calculation.

The report expands those source entries into explanatory sections. Expansion means explaining how the supplied values connect to plant planning, material flow, quality, safety and logistics; it does not change the source figures. Where a topic requires information that is not present in the source, the report identifies it as an item requiring validation rather than presenting a new number as fact.

This approach maintains traceability. A reviewer can compare the major tables in this document with the original presentation and identify the same cost ranges, employee categories and production assumptions. It also makes clear which parts are supplied project data and which parts are explanatory planning discussion.

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5. Industrial Context of Windmill Manufacturing

Windmill manufacturing is an integrated industrial activity involving structural fabrication, precision machining, composite processing, assembly, inspection and logistics. Large wind-turbine components cannot be treated like ordinary small manufactured products because component dimensions influence factory clearances, lifting arrangements, storage areas and transport routes.

The proposed facility therefore combines several manufacturing disciplines. Cutting and forming equipment supports fabrication; welding equipment joins structural parts; CNC equipment supports precision work; composite equipment supports blade manufacture; cranes and hydraulic equipment support handling; and testing equipment provides a final verification function.

The important planning principle is interdependence. A decision about blade size can affect mould dimensions, building length, curing space, crane reach, storage and transportation. Similarly, a decision to manufacture more components internally affects machine investment, operators, maintenance, inspection and working capital.

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6. Need for an Integrated Facility Study

A manufacturing facility requires more than a list of machines. The machines must have a place in the process, suitable utilities, operators, maintenance access, inspection points and material-flow connections. The supplied project therefore combines land, buildings, machinery, manpower, logistics and finance.

The integrated approach is especially relevant for large blades. A blade may require a large mould, composite reinforcement, resin systems, controlled processing, curing, trimming, inspection and safe handling. Once completed, the component must move from the production area to storage and then to its destination using suitable transport equipment.

The report consequently treats the company as a system. Each department contributes to the same production objective: engineering defines requirements, procurement provides materials and equipment, production converts inputs into components, quality verifies conformity, maintenance protects equipment availability, logistics moves materials and products, and management coordinates resources.

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7. Objectives

The central objective is to document the proposed windmill manufacturing facility as a coherent industrial system. Technical objectives include describing the land requirement, building areas, manufacturing machinery, blade production, assembly, testing, workforce, transportation and cost model.

A second objective is traceability of assumptions. The report keeps the 12-acre land concept, the selected ₹420 crore investment scenario, the 450–550 employee model, the ₹175 crore annual operating estimate and the 100-blade annual production assumption connected throughout the document.

A third objective is to identify the information needed for future refinement. Site surveys, detailed layouts, equipment quotations, process specifications, quality plans, route studies, environmental requirements and financial models would be needed to progress from a conceptual report to implementation-level documentation.

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8. Functional Requirements

The facility needs to receive and store raw materials, perform fabrication and machining, manufacture composite blade components, assemble and handle large parts, inspect and test products, store finished products and dispatch them. Administrative and technical functions such as engineering, quality, HR, finance and IT support these operations.

Each functional requirement creates physical and organizational consequences. Receiving requires unloading and inspection space; fabrication requires cutting, forming and welding areas; machining requires controlled equipment zones; composite manufacturing requires mould and process space; assembly requires lifting capacity; and dispatch requires a suitable route from finished-goods storage to the exit.

The requirements should ultimately be converted into measurable design criteria. For example, machine foundation loads, crane capacity, aisle widths, clear heights, storage capacity, electrical loads and inspection tolerances should be defined from approved engineering information rather than assumed from a conceptual table.

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9. Non-Functional Requirements

Safety, reliability, traceability, maintainability, scalability, quality consistency, efficient material handling and regulatory compliance are important non-functional requirements for the facility. These requirements influence the plant even when they are not directly visible in the production sequence.

Traceability is particularly important because materials, process settings and inspection results may need to be connected to a component.

Maintainability requires access around equipment, planned maintenance windows and spare-parts control. Scalability requires enough space and infrastructure flexibility to accommodate future changes without creating unsafe congestion.

Quality consistency depends on controlled procedures rather than final inspection alone. Safety similarly depends on engineering controls, procedures, training and supervision. These non-functional requirements should therefore be considered during layout and equipment selection rather than added after construction.

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10. Engineering Methodology

The supplied report uses a staged engineering sequence: requirement study, site study, plant planning, equipment planning, manpower planning, production and quality planning, logistics, cost modelling, web presentation and validation. This sequence provides a logical dependency between physical requirements and financial assumptions. The method begins with requirements because a machine list without a defined process can produce inappropriate capacity. Site planning follows because component size, vehicle movement and utilities affect land use. Equipment planning then establishes the principal manufacturing capability, followed by manpower and quality requirements.

The final cost model is therefore connected to the preceding sections. If machine quantities change, capital cost changes. If production capacity changes, raw-material use, staffing and transport can change. A controlled revision process should be used whenever an assumption is modified.

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11. Site Concept — Thoothukudi

The proposed company is conceptually located at Thoothukudi, Tamil Nadu. The supplied material emphasizes port-linked logistics and road connections toward wind-energy regions. For a large-component manufacturing facility, location suitability depends on more than distance: access roads, turning geometry, utilities, drainage, emergency access and future expansion are also relevant.

The site is presented as approximately 12 acres. This gross area must be interpreted carefully because roads, parking, open space, setbacks, buildings and utility zones all compete for land. The report therefore treats the 12-acre figure as the planning basis rather than a detailed site boundary.

Before implementation, a site-specific survey would be required. Such a study would verify land boundaries, ground levels, soil conditions, drainage, access, utility availability, permitted use and the feasibility of moving oversized components.

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12. Land Cost Analysis

The presentation gives a minimum land-price calculation of 12 acres $\times$ ₹12 crore per acre = ₹144 crore and a maximum calculation of 12 acres $\times$ ₹45 crore per acre = ₹540 crore. The financial model selects ₹300 crore as an average planning value.

The range demonstrates the sensitivity of total investment to land price. Because land is a major part of the selected ₹420 crore scenario, changes in the actual acquisition price would have a direct effect on the capital requirement. Transaction costs and site-development costs are not established by the supplied table and would need separate assessment.

The land figure should therefore be used as a scenario input. A detailed acquisition study would record the actual parcel, legal status, permitted use, access, survey, utility connections and associated transaction costs before a final investment figure is approved.

Source table from the supplied presentation

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13. Land Cost Table

The following table reproduces the land-cost information from the supplied presentation. It is followed by an interpretation so that the table is not isolated from the engineering discussion.

The ₹144 crore lower scenario is based on ₹12 crore per acre, while the ₹540 crore upper scenario is based on ₹45 crore per acre. The difference between the two values is ₹396 crore, showing that land-price uncertainty can materially influence the investment model.

The selected ₹300 crore average is a modelling choice in the supplied material. It should be retained as an assumption until supported by a current site-specific valuation.

Source table from the supplied presentation

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14. Building Area Planning

The presentation allocates 200,000 sq ft to the factory shed or production area, 20,000 sq ft to office and R&D, 100,000 sq ft to warehouse and storage, and 202,720 sq ft to roads, parking and open space. The combined area is 522,720 sq ft.

The production area is the largest built component because manufacturing requires space for machinery, work-in-progress, material handling and large assemblies. The warehouse provides separation between incoming materials and production, while office and R&D space supports engineering and administrative functions.

Roads and open space are also essential rather than residual. Large components and trailers require clear movement paths, turning areas and emergency access. A detailed site layout should therefore verify that building footprints and circulation can coexist within the stated land area.

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15. Building Cost Analysis

The presentation gives production-shed construction at ₹300–₹600 per sq ft, office and R&D at ₹1,500–₹2,500 per sq ft, warehouse and storage at ₹300–₹600 per sq ft, and roads/parking/open space at ₹50–₹150 per sq ft. The resulting building estimate is presented as ₹13–26 crore.

These rates are planning ranges from the supplied material. They should not be interpreted as current construction quotations because actual cost depends on structural design, foundations, floor loading, utilities, finishes, fire systems, site conditions and market conditions.

The building estimate should also be checked against the equipment list. Heavy machinery may require special foundations, while cranes require structural provisions and safe clearances. Composite processing may require controlled environmental conditions and suitable ventilation.

Source table from the supplied presentation

Section	Area	Cost per sq . ft	Total cost
Factory shed / (production)	200000	300 - 600	6 – 12 cr
Office & R&D	20000	1500 – 2500	3 – 5 cr
Warehouse & storage	100000	300 – 600	3 – 6 cr
Roads, parking, open space	202720	50 - 150	1- 3 cr

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16. Building Area and Cost Table

This page records the supplied building table in a report-ready format. The factory shed has the largest area allocation, while office and R&D has the highest cost per square foot range. Warehouse and storage provides significant support space for materials and finished components.

The roads, parking and open-space line is particularly important for a large-component facility. Transport planning cannot be separated from the building plan because finished blades may require specialized trailers and controlled movement paths.

The table should be treated as the baseline for later layout work. Any change to building area or cost should be reflected in the capital investment model.

Source table from the supplied presentation

Section	Area	Cost per sq . ft	Total cost
Factory shed / (production)	200000	300 - 600	6 – 12 cr
Office & R&D	20000	1500 – 2500	3 – 5 cr
Warehouse & storage	100000	300 – 600	3 – 6 cr
Roads, parking, open space	202720	50 - 150	1- 3 cr

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17. Factory Shed Engineering

The factory shed must accommodate machines, operators, work-in-progress and material-handling equipment. Clear height and floor capacity are important because the facility includes cranes, large moulds and heavy machinery.

The shed layout should provide access for maintenance and safe movement around machines. Equipment foundations should be designed using actual vendor data, including machine mass, dynamic effects and utility connections. Crane systems should be coordinated with structural design and lifting plans.

Lighting, ventilation, fire protection, electrical distribution and emergency exits are also part of the factory environment. These elements should be incorporated into the detailed design rather than treated as optional additions.

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18. Warehouse and Storage

Warehouse planning must distinguish raw materials, consumables, work-in-progress and finished products. Composite materials and resin systems may require controlled storage conditions and identification by batch. Metal materials require suitable protection against damage and contamination.

Storage locations should be linked to inventory records. A simple location-control method can identify the material, quantity, batch, receipt date and issue status. This supports production planning and traceability.

Finished blades require a different storage approach because of their dimensions. The storage area must allow safe movement, support, protection from damage and access for dispatch equipment. The warehouse plan should therefore be coordinated with transport routes.

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19. Utilities and Infrastructure

The proposed facility requires electrical power, water, drainage, lighting, compressed air where applicable, ventilation, fire protection and communication systems. Utility capacity should be derived from the actual selected equipment rather than only from the equipment names.

Utility reliability affects production continuity. A power interruption during a controlled composite process, machining operation or testing activity may create quality or safety concerns. Critical processes should therefore be assessed for backup or controlled shutdown requirements. Infrastructure planning also includes internal roads, parking, drainage and emergency access. The infrastructure design should be reviewed together with the site plan, machine layout and safety requirements.

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20. Machinery Overview

The supplied presentation groups machinery into general fabrication, precision machines, blade manufacturing, assembly and handling, and testing. The categories create a broad production capability rather than a single manufacturing process.

The listed machines include CNC plasma cutting, CNC laser cutting, press brake, plate rolling, submerged arc welding, robotic welding, CNC drilling, shot blasting and painting. Precision equipment includes gear hobbing, gear grinding, CNC lathe and CNC milling. Blade-related equipment includes an FRP blade mould and vacuum infusion system. Machine selection should ultimately consider capacity, accuracy, work envelope, tooling, utilities, operator requirements, maintenance support and lifecycle cost. The supplied price ranges are retained as planning values.

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21. General Fabrication Machinery

General fabrication begins with material preparation and continues through cutting, forming, drilling, welding, surface preparation and painting. The supplied machine list supports these operations through CNC plasma, CNC laser, press brake, plate rolling, welding, drilling, shot blasting and painting systems.

The sequence should be organized to minimize unnecessary handling. Cut components should move logically to forming or drilling, then to welding and surface preparation. Inspection points should be inserted where dimensions or weld quality can be verified before subsequent operations conceal defects.

The machine list also creates manpower requirements. Operators need process-specific training, while maintenance staff need access to machine documentation, consumables and spare parts.

Source table from the supplied presentation

Machine	Code	Price Range
CNC Plasma Cutting Machine	CNC-PLASMA	₹5 lakh – ₹30 lakh
CNC Laser Cutting Machine	CNC-LASER	₹20 lakh – ₹1 Crore
CNC Press Brake	CNC-PB	₹5 lakh – ₹80 lakh
Plate Rolling Machine (4 roll)	4R-ROLL	₹10 lakh – ₹50 lakh
Submerged Arc Welding	SAW	₹1 lakh – ₹5 lakh
Robotic Welding System	RWS	₹20 lakh – ₹2 Crore
CNC Drilling Machine	CNC-DRILL	₹5 lakh – ₹25 lakh
Shot Blasting Machine	SBM	₹10 lakh – ₹1 Crore
Painting Booth	PBS	₹5 lakh – ₹30 lakh

System		
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22. Fabrication Machine Table

The following table reproduces the principal fabrication machines and price ranges from the supplied PPT. The price ranges are presented exactly as planning ranges rather than as verified procurement quotations.

The table is useful for comparing the relative equipment scope. It also shows why the total machine investment cannot be calculated simply by adding the lower or upper values without considering installation, tooling, auxiliary systems and equipment not individually priced in the table.

A procurement-stage equipment register should add supplier, lead time, utility requirement, foundation, warranty, training, service support and acceptance-test information.

Source table from the supplied presentation

Machine	Code	Price Range
CNC Plasma Cutting Machine	CNC-PLASMA	₹5 lakh – ₹30 lakh
CNC Laser Cutting Machine	CNC-LASER	₹20 lakh – ₹1 Crore
CNC Press Brake	CNC-PB	₹5 lakh – ₹80 lakh
Plate Rolling Machine (4 roll)	4R-ROLL	₹10 lakh – ₹50 lakh
Submerged Arc Welding	SAW	₹1 lakh – ₹5 lakh
Robotic Welding System	RWS	₹20 lakh – ₹2 Crore
CNC Drilling Machine	CNC-DRILL	₹5 lakh – ₹25 lakh
Shot Blasting Machine	SBM	₹10 lakh – ₹1 Crore
Painting Booth System	PBS	₹5 lakh – ₹30 lakh

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23. Precision and Gear Machinery

Precision manufacturing in the supplied concept includes gear hobbing, gear grinding, CNC turning and CNC milling. These machines support components requiring controlled geometry and dimensional accuracy.

Precision equipment should be located where environmental and process conditions can be managed. Inspection capability must be matched to the required tolerances, and measurement equipment should remain calibrated. Workholding and tooling are also important because machine capability alone does not guarantee part accuracy.

The machine list should be linked to the actual parts intended for internal manufacture. If certain components are outsourced, machine capacity and capital investment could be reduced; if more components are brought in-house, additional tooling and operators may be required.

Source table from the supplied presentation

 Machine	 Code	 Price
Gear Hobbing Machine	GH	₹5 lakh – ₹50 lakh
Gear Grinding Machine	GG	₹20 lakh – ₹2 Crore
CNC Lathe Machine	CNC-LATHE	₹5 lakh – ₹40 lakh
CNC Milling Machine	CNC-MILL	₹10 lakh – ₹50 lakh

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24. Blade Manufacturing Equipment

The presentation identifies an FRP Blade Mould System and a Vacuum Infusion System. These are central to the composite blade manufacturing concept. The mould establishes the component geometry, while the infusion system supports the resin-processing stage. The blade manufacturing area must provide adequate length, handling clearance and controlled process conditions. Mould maintenance is important because dimensional or surface defects can transfer to the product. Resin and reinforcement materials require identification and controlled storage.

The equipment should be supported by documented process parameters, inspection checkpoints and operator training. Exact resin systems, curing conditions and structural acceptance criteria are not specified by the supplied presentation and therefore require separate engineering definition.

Source table from the supplied presentation

 Machine	 Code	 Price
FRP Blade Mould System	FRP-MOULD	₹50 lakh – ₹5 Crore
Vacuum Infusion System	VIS	₹10 lakh – ₹1 Crore

Engineering interpretation and control points

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25. Assembly and Handling Equipment

The assembly and handling category includes a hydraulic press, EOT crane in the 10–20 tonne range and assembly line system. These systems support movement, positioning and joining of large components.

Handling equipment is a safety-critical part of the facility. The actual crane capacity and configuration must be verified against the heaviest intended load, centre of gravity, lifting accessories and operating conditions. Lifting plans should define exclusion zones and communication methods.

Assembly fixtures should be designed to hold components without damage and to provide repeatable positioning. The assembly area should also be accessible for inspection and final testing.

Source table from the supplied presentation

 Machine	 Code	 Price
Hydraulic Press	HP	₹5 lakh – ₹50 lakh
EOT Crane (10–20 ton)	EOT	₹5 lakh – ₹25 lakh
Assembly Line System	ALS	₹10 lakh – ₹1 Crore

Engineering interpretation and control points

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26. Testing Equipment

The supplied presentation identifies a Wind Turbine Testing System with a planning price range of ₹1–10 crore. Testing is important because the finished product must be compared against defined requirements before release.

The exact test scope is not specified in the presentation. A detailed test plan would define the item under test, method, instrumentation, calibration status, test conditions, acceptance criteria and reporting format. Test records should remain traceable to the component or assembly serial number.

Testing should not be treated as a single final activity. Incoming inspection, in-process checks, dimensional verification and final testing form a chain of evidence supporting product quality.

Source table from the supplied presentation

 Machine	 Code	 Price
Wind Turbine Testing System	WT-TEST	₹1 Crore – ₹10 Crore

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27. Total Machine Investment

The presentation gives large-scale machine investment as ₹80–150+ crore and selects ₹100 crore in the total investment model. Small-scale and medium-scale scenarios are also shown as ₹20–40 crore and ₹40–80 crore respectively.

The selected ₹100 crore figure represents the scale chosen for the concept. It should be reconciled with the individual machine list, installation requirements, tooling, utilities, civil foundations, commissioning and contingency before procurement.

The machine investment is one of the three major capital categories in the selected ₹420 crore scenario, alongside ₹300 crore land and ₹20 crore building. Changes in machine scope therefore directly affect the total capital requirement.

Source table from the supplied presentation

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28. Machine Investment Table

This section summarizes the scale-based machine investment presented in the PPT: small scale ₹20–40 crore, medium scale ₹40–80 crore, and large scale ₹80–150+ crore. The selected project model uses ₹100 crore. The table provides a scenario range rather than a detailed bill of quantities. A future procurement plan should convert it into a machine-by-machine schedule, including quantity, price, installation, tooling, taxes, freight and commissioning.

This distinction is important for financial control because a headline machine budget can hide significant supporting expenditure. The detailed machine tables earlier in the report should therefore be retained as technical evidence alongside the scenario total.

Source table from the supplied presentation

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29. Workforce Overview

The supplied presentation estimates 450–550 employees. The summary groups the workforce into management, engineers, technicians and operators, office staff and support staff. Engineers and production personnel form a major portion because the facility combines manufacturing, maintenance, quality and technical functions.

The workforce plan should be aligned with the operating schedule. A multi-shift arrangement changes staffing requirements, maintenance coverage and supervision. Competency requirements should be defined for machine operation, composite processing, lifting, inspection, maintenance and safety.

The total headcount is a planning range. The final organization would depend on actual production capacity, automation level, outsourcing decisions, shift pattern and support requirements.

Source table from the supplied presentation

Category	Employees
Management	10 – 15
Engineers (All types)	120 – 180
Technicians & Operators	150 – 220
Office Staff (HR, Finance, IT)	40 – 60
Support Staff	100 – 150
Total	450 – 550 Employees

Engineering interpretation and control points

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30. Employee Summary Table

The PPT summary gives 10–15 management employees, 120–180 engineers, 150–220 technicians and operators, 40–60 office staff and 100–150 support staff, for a total of 450–550 employees.

The range allows the organization to scale within the conceptual model. Engineering and technical positions are needed because the facility contains specialized machines and composite processes. Support staff cover logistics, administration, security and general activities.

A detailed manpower budget should reconcile individual roles with these group totals and then connect salaries to the annual operating-cost model. Recruitment timing should also follow commissioning and production ramp-up.

Source table from the supplied presentation

Category	Employees
Management	10 – 15
Engineers (All types)	120 – 180
Technicians & Operators	150 – 220
Office Staff (HR, Finance, IT)	40 – 60
Support Staff	100 – 150
Total	450 – 550 Employees

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31. Management and Project Roles

The detailed employee table lists a CEO or Managing Director, General Managers and Project Managers. These roles provide overall direction, departmental supervision and coordination of wind-related projects.

Management responsibilities should include resource planning, quality governance, safety leadership, financial control and schedule coordination. Project roles should coordinate engineering, procurement, installation and delivery activities where the manufacturing organization supports external wind projects.

Clear responsibility boundaries reduce duplication. A responsibility matrix can identify who approves engineering changes, production release, quality disposition, procurement and dispatch.

Source table from the supplied presentation

S.No	Job Role	Department	No. of Employees	Salary (Per Month ₹)	Salary (Per Year ₹)	Job Description
1	CEO / Managing Director	Management	1	3,00,000 – 6,00,000	36L – 72L	Overall company control
2	General Manager	Management	2 – 3	1,50,000 – 3,00,000	18L – 36L	Department supervision
3	Project Manager	Project	5 – 8	80,000 – 1,50,000	10L – 18L	Handle wind projects

Engineering interpretation and control points

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32. Engineering Roles

The PPT identifies mechanical engineers, electrical engineers, R&D engineers and production engineers. Mechanical engineering supports turbine-part design and manufacturing; electrical engineering supports power systems; R&D supports innovation; production engineering supports process execution.

These roles interact continuously. A change in component geometry may affect machining, electrical integration, assembly and testing.

Production engineering converts design information into practical process steps and coordinates with quality and maintenance.

Engineering records should use controlled revisions. Drawings, specifications, process sheets and inspection criteria should be traceable so that production does not rely on obsolete information.

Source table from the supplied presentation

4	Mechanical Engineers	Engineering	40 – 60	30,000 – 80,000	4L – 10L	Design turbine parts
5	Electrical Engineers	Engineering	30 – 50	35,000 – 90,000	5L – 11L	Power systems

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33. Operators and Technicians

Machine operators are listed at 80–120 employees, while wind-turbine technicians are listed at 70–100. These roles are directly connected to production and maintenance activities.

Operators require machine-specific competency, safe working practices and understanding of process limits. Technicians require knowledge of equipment installation, maintenance, fault finding and field-related work where applicable. Both groups contribute to operational reliability. Training records should show the equipment or task authorized, assessment status and refresher requirements. Safety-critical work should not be assigned solely on the basis of attendance at training.

Source table from the supplied presentation

6	R&D Engineers	Research	10 – 20	50,000 – 1,20,000	6L – 15L	Innovation & development
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34. Site, Quality and Safety Roles

The presentation lists site engineers, quality engineers and safety officers. Site engineers support supervision; quality engineers support inspection and process control; safety officers support safe working practices.

These functions provide independent checks around production. Quality personnel should have access to inspection records and authority to place affected material on hold when requirements are not met. Safety personnel should monitor hazards, emergency preparedness and compliance with procedures.

The organization should ensure that production pressure does not remove necessary quality or safety controls. Escalation paths should be defined before problems occur.

Source table from the supplied presentation

7	Production Engineers	Manufacturing	30 – 50	30,000 – 75,000	4L – 9L	Production process
8	Machine Operators	Production	80 – 120	18,000 – 35,000	2L – 4L	Operate machines
9	Wind Turbine Technicians	Maintenance	70 – 100	20,000 – 50,000	2.5L – 6L	Install & repair turbines

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35. Logistics, HR, Finance and IT Roles

The employee tables include store and inventory staff, supply chain managers, HR executives, finance and accounts, IT support, sales and marketing, administration, security and helpers/workers.

These support functions keep the manufacturing system operational. Inventory staff control material movement, supply chain staff coordinate suppliers and logistics, HR manages workforce processes, finance controls budgets, and IT supports information systems. Sales and marketing connect the manufacturing capability to business demand.

The range of support roles demonstrates that manufacturing is broader than the production floor. Administrative and digital systems must therefore be planned alongside machines and buildings.

Source table from the supplied presentation

9	Wind Turbine Technicians	Maintenance	70 – 100	20,000 – 50,000	2.5L – 6L	Install & repair turbines
10	Site Engineers	Project Site	25 – 40	25,000 – 60,000	3L – 7L	Site supervision
11	Quality Engineers	Quality	15 – 25	30,000 – 70,000	4L – 9L	Quality checking
12	Safety Officers	Safety	10 – 15	25,000 – 60,000	3L – 7L	Ensure safety rules
13	Store & Inventory Staff	Logistics	20 – 30	20,000 – 40,000	2.5L – 5L	Material handling
14	Supply Chain Manager	Logistics	5 – 8	60,000 – 1,30,000	7L – 16L	Logistics control
15	HR Executives	HR	10 – 15	20,000 – 60,000	3L – 7L	Recruitment & staff care
16	Finance &	Finance	10 –	30,000 –	5L –	Accounts &

	Accounts		20	90,000	12L	budgeting
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36. Salary Expense

The PPT gives an estimated monthly salary expense of ₹1.8–3 crore and annual salary expense of ₹20–35 crore. The selected operating-cost model uses ₹25 crore per year.

The range reflects the broad 450–550 employee organization and the different salary bands in the detailed employee table. A detailed payroll model would calculate headcount × monthly salary for each role and then add applicable benefits, contributions, overtime, contractor costs and other employment expenses where relevant.

The ₹25 crore selected value should therefore be treated as a planning assumption. It is not derived in the PPT from a complete role-by-role payroll reconciliation.

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37. Blade Manufacturing Context

The blade section of the presentation covers several material categories, from fiberglass and carbon composites to hybrid materials, aluminium, plastic/nylon, experimental bio-materials and segmented modular blades.

The categories illustrate the range of blade concepts represented in the presentation. Material choice affects structural performance, weight, manufacturing process, tooling, cost, handling and transport. Composite materials also require controlled processing and traceability.

The report preserves the supplied ranges without treating every listed category as a confirmed product line of the proposed facility. Actual product selection would require detailed engineering, customer requirements and validation.

Source table from the supplied presentation

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38. Blade Material Table — Part I

The supplied table lists fiberglass blades at 30–80 m with a manufacturing cost of ₹1.2–3.5 crore per blade; carbon-fibre composite blades at 50–100+ m with ₹3–8 crore; hybrid glass plus carbon at 60–100 m with ₹2.5–6 crore; and aluminium blades at 1–5 m with ₹15,000–₹2 lakh.

These figures are presentation values. The very different size and cost ranges demonstrate that “blade” is not a single manufacturing category. Production planning must identify the intended blade geometry before determining mould size, labour, cycle time and transport requirements. Cost also depends on material system, process route, labour, tooling, inspection and yield. Therefore the presentation values should remain labelled as approximate manufacturing-cost ranges.

Source table from the supplied presentation

Blade Type (Material)	Typical Length (Height)	Manufacturing Cost (per blade)
Fiberglass (Glass Fiber Composite)	30 m – 80 m	₹1.2 Cr – ₹3.5 Cr (\$150k–\$400k)
Carbon Fiber Composite	50 m – 100+ m	₹3 Cr – ₹8 Cr (\$400k–\$1M+)
Hybrid (Glass + Carbon Fiber)	60 m – 100 m	₹2.5 Cr – ₹6 Cr

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39. Blade Material Table — Part II

The PPT adds plastic or nylon blades at 0.5–2 m and ₹1,000–₹10,000, wood or experimental bio-material blades at 10–40 m and ₹50 lakh–₹2 crore, and segmented modular blades at 70–115 m and ₹3–10 crore.

The segmented category is particularly relevant to logistics because modular construction can change how a large blade is handled and transported. The supplied presentation identifies these as one of the blade categories rather than providing a detailed process design.

A future product definition should specify material, dimensions, structural design, manufacturing method, inspection requirements and customer acceptance criteria. Until then, the table remains a conceptual comparison.

Source table from the supplied presentation

Aluminium Blades	1 m – 5 m	₹15,000 – ₹2 Lakh
Plastic / Nylon Blades	0.5 m – 2 m	₹1,000 – ₹10,000
Wood / Experimental Bio-material	10 m – 40 m	₹50 Lakh – ₹2 Cr
Segmented Blades (Modular)	70 m – 115 m	₹3 Cr – ₹10 Cr

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40. Blade Manufacturing Sequence

A conceptual blade process begins with design and tooling definition, followed by mould preparation, material lay-up, vacuum or resin processing, curing, demoulding, trimming, finishing, inspection, identification and storage.

The sequence contains several quality-sensitive stages. Mould condition influences geometry; lay-up influences material distribution; resin processing influences consolidation; curing influences the final material state; and trimming influences dimensions and surface condition.

The process should use controlled work instructions and records. Material batch, operator, process parameters and inspection results should be linked to the blade identification number so that quality evidence remains traceable.

Engineering interpretation and control points

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41. Mould Preparation and Lay-up

Mould preparation establishes the surface on which the blade component is formed. Cleaning, release preparation and dimensional checks are therefore important before material placement begins.

Lay-up places reinforcement materials according to the defined process. Consistency in orientation, overlap, placement and identification is important because the final structure depends on the material arrangement. The supplied presentation does not provide a detailed laminate schedule, so this report does not invent one.

A practical control record would identify the mould, material batches, work instruction revision, operator, start and completion time, inspection checkpoints and any deviations requiring engineering review.

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42. Vacuum Infusion and Curing

The presentation identifies a Vacuum Infusion System. In a conceptual process, the system supports resin movement through reinforcement under controlled conditions. Vacuum integrity and process monitoring are important because defects can arise when the process is not controlled.

Curing must follow the approved material system and process specification. Exact temperatures, times, resin ratios and vacuum values are not supplied in the PPT and therefore require qualified process development before production.

After curing, the component can proceed to demoulding and finishing. The transition should include checks for surface condition, dimensional conformity and visible defects before the part moves to later stages.

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43. Blade Finishing and Inspection

After demoulding, a blade component may require trimming, edge finishing, surface treatment and dimensional inspection. The inspection plan should identify critical dimensions and surfaces and define the method used to verify them.

Potential manufacturing concerns include voids, dry areas, wrinkles, delamination, bonding defects, dimensional deviation and surface damage. The supplied material does not define acceptance limits, so the report treats these as quality-control considerations rather than product-specific defect limits.

Inspection results should be recorded against the blade identification number. If a non-conformance is detected, the affected component should be controlled until an approved disposition is made.

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44. Blade Handling and Storage

Large blades require controlled handling because their dimensions make them vulnerable to impact and bending during movement. The storage arrangement should support the component appropriately and maintain access for inspection and dispatch.

Handling plans should define lifting points, accessories, load distribution and communication. The actual lifting equipment must be selected using verified load and centre-of-gravity data. The presentation provides handling equipment categories but does not provide detailed blade mass calculations.

Storage should also protect finished surfaces and provide a clear path to the transport area. The storage plan should be coordinated with dispatch schedules so that unnecessary rehandling is avoided.

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45. Production Flow

The conceptual production flow is receiving, incoming inspection, raw-material storage, fabrication or machining, blade/composite processing, assembly, testing, finished-goods storage and dispatch.

This flow should be arranged to minimize backtracking. Inspection holds should be located where they can stop defects before they move downstream. Work-in-progress identification should remain visible throughout the process.

Production control connects the flow to the annual 100-blade assumption. Actual capacity would depend on cycle times, mould availability, curing time, machine availability, labour, quality yield and customer specifications. The 100-blade value is therefore an illustrative planning capacity.

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46. Production Scheduling and Bottlenecks

Production scheduling should coordinate material availability, machine capacity, composite processing, curing, assembly, inspection and dispatch. The slowest or most constrained operation can determine overall throughput.

If a mould or curing stage limits the process, increasing unrelated machine capacity may not increase final output. Similarly, high machine availability is not useful if material inspection or quality release creates a downstream hold.

A future production-control system could track planned quantity, actual quantity, work-in-progress, downtime, quality holds and dispatch status. These records would support capacity analysis and continuous improvement.

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47. Assembly Planning

Assembly combines components using controlled procedures, fixtures and appropriate tools. The supplied presentation includes an assembly line system, hydraulic press and EOT crane.

Assembly planning should define sequence, torque or joining requirements where applicable, inspection points and access for maintenance or rework. Fixtures should provide repeatable positioning and avoid damaging finished surfaces.

Final assembly should be linked to quality release. Components with unresolved inspection issues should not be incorporated into the final product without an approved disposition. Identification and documentation should remain connected through the assembly stage.

Engineering interpretation and control points

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48. Testing and Release

Testing provides evidence that the completed product meets defined requirements. The supplied project identifies a wind-turbine testing system, while the detailed acceptance criteria are not specified.

A release process should therefore include review of material certificates where required, in-process inspection, dimensional checks, assembly records, calibration status and final test results. The responsible authority should sign or electronically approve release.

Test records should be retained with product identification and revision information. This creates a traceable record that can support customer documentation and later investigation if a field issue occurs.

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49. Quality Assurance Framework

Quality assurance establishes controlled processes intended to prevent defects, while quality control verifies that outputs meet requirements. The supplied report proposes controls across incoming materials, fabrication, machining, composite processing, assembly, final inspection and dispatch.

The strongest quality system places checks at meaningful points in the process. Incoming material checks prevent incorrect inputs; in-process checks prevent defects from moving forward; final inspection verifies the completed product.

Document control is part of quality. Work instructions, drawings, inspection forms and process specifications should have controlled revisions. Obsolete documents should be removed from active work areas.

Engineering interpretation and control points

The information on this page should be read together with the adjacent process, layout, quality and cost sections. A change in the underlying assumption should be recorded with its source, revision, owner and date, then checked for its effect on connected requirements. For implementation, the responsible function should define the required input, the acceptance criterion and the evidence needed to demonstrate completion. Where the supplied presentation does not define a value, the value should remain an open engineering or commercial decision rather than being treated as verified data.

A controlled record can include the applicable drawing or specification, equipment or material identification, inspection result, approval status and follow-up action. This provides continuity between the written report, the presentation and the digital information layer. It also helps reviewers distinguish source data from future implementation decisions.

50. Quality Control Table

The source report identifies the following control stages: incoming material identity and condition; fabrication dimensions, weld quality and surface condition; machining dimensions and tolerances; composite lay-up, infusion and cure; assembly fitment and procedure compliance; final dimensional, visual and functional inspection; and dispatch identification and documentation.

The table is useful because it shows that quality is distributed across the process. Each stage should have a defined requirement, method, acceptance criterion and record.

Where a value is not supplied, the detailed quality plan should be created from approved engineering documentation and customer requirements.

Engineering interpretation and control points

The information on this page should be read together with the adjacent process, layout, quality and cost sections. A change in the underlying assumption should be recorded with its source, revision, owner and date, then checked for its effect on connected requirements. For implementation, the responsible function should define the required input, the acceptance criterion and the evidence needed to demonstrate completion. Where the supplied presentation does not define a value, the value should remain an open engineering or commercial decision rather than being treated as verified data.

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